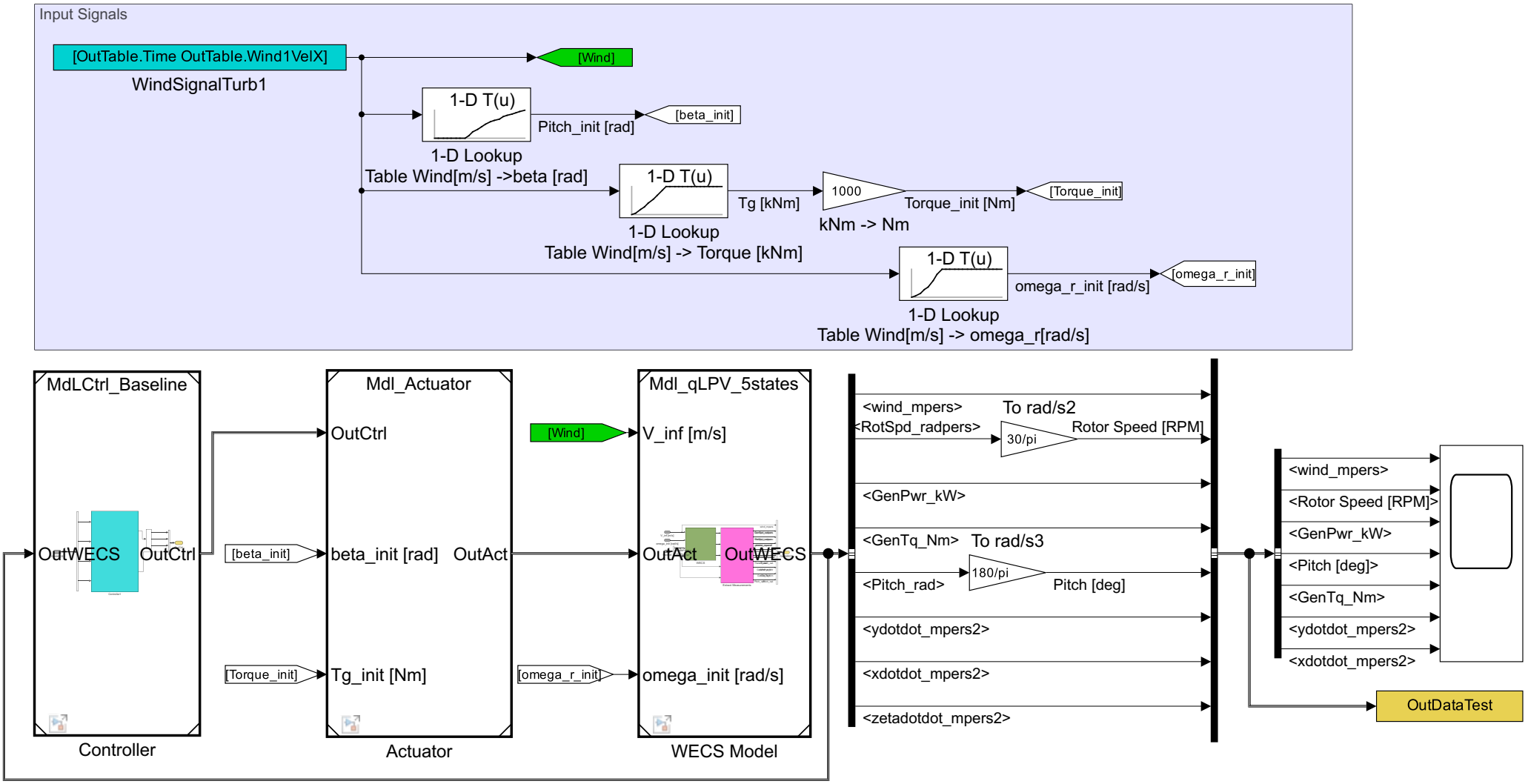

Input Signals
[OutTable.Time OutTable.Wind1VelX]
WindSignalTurb1
[Wind]
1-D T(u)
Pitch_init [rad]
[beta_init]
1-D Lookup
Table Wind[m/s] ->beta [rad]
1-D T(u)
Tg [kNm]
1000
Torque_init [Nm]
[Torque_init]
kNm -> Nm
1-D Lookup
Table Wind[m/s] -> Torque [kNm]
1-D T(u)
omega_r_init [rad/s]
[omega_r_init]
1-D Lookup
Table Wind[m/s] -> omega_r[rad/s]
MdLCtrl_Baseline
OutWECS
OutCtrl
Controller
Mdl_Actuator
OutCtrl
[beta_init]
beta_init [rad]
OutAct
[Torque_init]
Tg_init [Nm]
Actuator
[Wind]
[omega_r_init]
Mdl_qLPV_5states
V_inf [m/s]
OutAct
OutWECS
omega_init [rad/s]
WECS Model
<wind_mpers>
<RotSpd_radpers>
To rad/s2
30/pi
Rotor Speed [RPM]
<GenPwr_kW>
<GenTq_Nm>
To rad/s3
<Pitch_rad>
180/pi
Pitch [deg]
<ydotdot_mpers2>
<xdotdot_mpers2>
<zetadotdot_mpers2>
<wind_mpers>
<Rotor Speed [RPM]>
<GenPwr_kW>
<Pitch [deg]>
<GenTq_Nm>
<ydotdot_mpers2>
<xdotdot_mpers2>
OutDataTest